Janindu Bandara

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About

- Master's candidate in **Electrical Communication Engineering** at the **University of Kassel**, with a background spanning *embedded systems*, *digital hardware design*, and *signal processing*.
- Hands-on experience programming **microcontrollers** (ATmega328P, ESP32, NodeMCU ESP8266) and designing **digital ICs** in *Verilog HDL*, verified through FPGA implementation and simulation testbenches.
- Conducted master's thesis research in collaboration with **Rohde & Schwarz GmbH & Co. KG**, developing and validating a large-scale MATLAB simulation framework for constrained 5G waveform optimization, reinforcing rigorous design, verification, and documentation practices in a professional engineering environment.
- Practical exposure to *robotics and sensor integration* (Webots, GPS-based navigation), *IoT hardware*, and *ECAD tools* (OrCAD); proficient in *C/C++*, *MATLAB*, and *Python*.

Experience

Master Thesis Student — *München, Germany*
Rohde & Schwarz GmbH & Co. KG — *Oct 2025 – April 2026*

- Designed **FDSS** pulse filters with **symmetric spectral extension (SE)** for DFT-s-OFDM waveforms, recovering a $\sim$2 dB **BER** penalty caused by per-subcarrier amplitude variation under **MMSE** equalization.
- Performed constrained waveform optimization over a **630-configuration parameter grid** (truncation factor, roll-off, extension ratio), balancing **PAPR**, **ACLR**, and throughput against **3GPP Power Class 3** limits; identified two **BER-optimal** operating regimes across a 0–10 dB SNR range.
- Benchmarked optimized **FDSS** waveform against **CP-OFDM**, **DFT-s-OFDM**, and **OTFS** across **AWGN**, **TDL-D** (indoor, 3 km/h), and **TDL-C** (highway, 120 km/h) 3GPP channel models; achieved **PAPR** as low as 2.87 dB while satisfying **ACLR** $\geq$ 30 dBc.
- **Tools & Systems Exposure:** *MATLAB (5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox)*

Temporary Demonstrator — *Colombo, Sri Lanka*
University of Sri Jayewardenepura — *Jan 2022 – June 2023*

- Worked in the Department of **Electrical and Electronic Engineering** at the University of Sri Jayewardenepura, assisting with *lab preparation, practical design, teaching support, evaluations, and workshops.*
- Gained hands-on experience across key areas including *electronics, communication systems, embedded systems, signal processing, automation, IoT, and power systems.*
- Programmed **ATmega328P** *and* **ESP32** *microcontrollers, simulated logic circuits, and configured static/routing protocols,* **VLANs**, *and* **DHCP/DNS** *on* **Cisco** *hardware.*
- **Tools & Systems Exposure:** *Arduino UNO, ESP32, Raspberry Pi, Atmel Studio, Proteus, Cisco Packet Tracer, NI Multisim, DIALux, Wireshark, VLAN, TCP/IP*

Freelance Engineer — *Remote*
Self-Employed (Fiverr) — *Sep 2022 – Aug 2023*

- Developed a dynamic web-based **IoT Visualization Interface** for real-time sensor data display on **Google Maps**, integrating **MQTT** and **Node-RED** for live map updates and marker control.
- **Tools Used:** *Node-RED, Google Maps API, JavaScript, MQTT, Docker*
- Designed a coordinated system integrating a differential drive **mobile robot** and a **3-DoF RRR manip-**

ulator for autonomous object transfer, implementing **GPS**-based navigation and **inverse kinematics** for pick-and-place tasks.

- **Tools Used:** *Webots, MATLAB (Robotics System Toolbox)*

Telecommunication Engineer Intern — *Colombo, Sri Lanka*
Sri Lanka Telecom PLC — *Feb 2020 – Aug 2020*

- Worked in the *Planning, Maintenance,* and *Network (Switching and Transmission) divisions.*
- Gained hands-on experience in *fiber splicing,* **OTDR**-*based verification, and fault diagnosis.*
- Assisted with *fiber splicing (fusion/mechanical)*, **ADSL/FTTH** *router configuration*, as well as the **WDM/MPLS** *Layer 2/3* **VPN** *provisioning*, *maintenance of transmission* and *switching equipment, including* **MSAN** *and* **eNodeB** *infrastructures.*
- **Tools & Systems Exposure:** *Fusion Splicer (Sumitomo T-400S), OTDR, Power Meter (FPM-300), MSAN, Clarity OSS, TCP/IP, MPLS, NGN*

Education

University of Kassel — *Kassel, Germany*
Master of Science in Electrical Communication Engineering — *Oct 2023 – Present*

- **Current Grade:** 1.68
- **Thesis:** Pulse Design for Pre-6G Mobile Radio Systems
- **Selected Coursework:** *Mobile Radio, Digital Communications, Information Theory & Coding, Signal Detection and Estimation, Simulation of Digital Communication Systems Using MATLAB, MAC Protocols in Wireless Communications,*

University of Sri Jayewardenepura — *Colombo, Sri Lanka*
Bachelor of Science of Engineering (Honours) in Electrical and Electronics (minor in Telecommunication Engineering) — *Nov 2016 – Jun 2021*

- **Grade :** 3.61 (out of 4.0) | equivalent to 1.5 in German Education System
- Second Class Honours (Upper Division)
- **Thesis:** Versatile Indoor Positioning System using Wi-Fi and Deep Learning
- **Selected Coursework:** *Embedded Systems, Digital Electronics, Data Communication, Wireless Communication, Signal Processing, IoT, Communications, Machine Learning, Image Processing, Next Generation Networks, Robotics, Mobile Application Development,*

Technical Skills

Programming Languages: *C++, Python, MATLAB, Java, JavaScript, Verilog*

Databases: *SQLite, MySQL*

Software / Tools: *Android Studio, Arduino IDE, Atmel Studio, AutoCAD, Cisco Packet Tracer, DIALux, Docker, LaTeX, NI Multisim, Node-RED, Octave, OrCAD, Proteus, Vivado, Webots, Wireshark*

Projects

Pulse Design for Pre-6G Mobile Radio Systems — *Oct 2025 – Jun 2026*

- Implemented link-level simulations of four candidate waveforms (CP-OFDM, DFT-s-OFDM, FDSS-enhanced DFT-s-OFDM, OTFS) across AWGN and 3GPP fading channels (TDL-D, TDL-C) using **MMSE equalization** under a perfect **channel state information (CSI)** assumption.
- Validated optimal FDSS design over 5×10^6 simulation slots per configuration; FDSS achieved **PAPR** 5.81 dB below CP-OFDM baseline while satisfying **ACLR** ≥ 30 dBc and **throughput** ≥ 10 Mbps.
- **Tools Used:** *MATLAB (5G NR Toolbox, Communications Toolbox, Signal Processing Toolbox, Parallel Computing Toolbox)*

SAE-CNN Based Indoor Positioning System — *Apr 2025 – Aug 2025*

- A deep learning-based indoor positioning system using Wi-Fi fingerprints, featuring a **stacked autoencoder (SAE)** for feature compression and a **1D-CNN** for classification.

- Applied powered normalization and uniform spatial sampling to improve signal stability and model generalization across indoor environments.
- **Tools Used:** *MATLAB (Deep Learning Toolbox), UJIIndoorLoc, and Tampere datasets*

Versatile Indoor Positioning System using Wi-Fi and Deep Learning *Oct 2020 – Jun 2021*

- A cost-effective indoor positioning system using **Wi-Fi RSSI** fingerprinting fused with **accelerometer-**derived displacement features, evaluated on self-collected data across two real rooms.
- Compared **KNN** and a **shallow ANN** with 5-fold cross-validation; accelerometer fusion improved KNN position accuracy by up to 35 percentage points (Room A: $39.0\% \rightarrow 74.4\%$); best result: KNN ($k = 5$), 72.77 cm mean error, 78.7% within 100 cm.
- **Tools Used:** *Python (scikit-learn, TensorFlow/Keras)*

Autonomous Search Rescue System *Oct 2020 – Feb 2021*

- Proposed a conceptual, cost-effective **autonomous search and rescue** system architecture to improve marine search and rescue operations, particularly for drowning victims, as part of an academic systems design study.
- Outlined a coordinated architecture in which a **UAV** tracks the victim, deploys a flotation device, and guides an **autonomous** rescue vessel, with **GPS** technology facilitating communication between components.

Gesture-Controlled Smart Room Kit *Dec 2018 – Mar 2019*

- A cost-effective solution to convert conventional electric appliances into smart devices, integrating **Wi-Fi** technology within the **smart home** concept.
- This product allows users to control existing home appliances via a smartphone, eliminating the need to replace entire appliances.
- **Tools Used:** *NodeMCU (ESP8266), Android Studio*

Root-Finding IC Design *Nov 2019 – Jan 2020*

- A digital IC designed for root-finding operations using the **IEEE-754 16-bit** floating-point format, featuring a **floating point unit (FPU)** and **master algorithm unit (MAU)**.
- Implemented arithmetic operations and developed hardware logic for **ln(x)** approximation using a 5th-order **Chebyshev polynomial** (Horner method evaluation), including normalization and internal data flow control.
- **Tools Used:** *Verilog HDL*

Language Skills

English: Fluent (C1)

German: Elementary (A2)

Sinhala: Native

Interests

Embedded Systems | **IoT** | **Robotics and Automation** | **Wireless Communication** | **Deep Learning** | **Machine Learning**

Declaration

I herewith declare that the details provided above are true and accurate to the best of my knowledge.

Ingolstadt, July 2, 2026

Janindu Bandara